HUNT-SAGE · THREAT HUNTING CHEAT SHEET

Sysmon Event IDs for Hunters

What each Sysmon event gives you and the techniques it exposes.

github.com/iamjammie/hunt-sage · iamjammie.github.io/hunt-sage · Built 2026-10-31

Sysmon (Microsoft Sysinternals) logs to `Microsoft-Windows-Sysmon/Operational`. Start from a maintained community configuration (for example SwiftOnSecurity's `sysmon-config` or Olaf Hartong's `sysmon-modular`) and tune for your environment. See Labs.

ID	Event	Key fields	Hunting use / techniques
1	Process creation	`Image`, `CommandLine`, `ParentImage`, `ParentCommandLine`, `User`, `Hashes`, `OriginalFileName`, `IntegrityLevel`, `ProcessGuid`	The core: execution chains, LOLBins, renamed binaries (T1059, T1218, T1036)
2	File creation time changed	`TargetFilename`, `CreationUtcTime`, `PreviousCreationUtcTime`	Timestomping (T1070.006)
3	Network connection	`Image`, `DestinationIp/Hostname/Port`, `Initiated`	C2, exfil, lateral (T1071, T1021); *high volume: filter*
4	Sysmon state changed		Service stopped = tamper (T1562)
5	Process terminated		Short-lived tooling
6	Driver loaded	`ImageLoaded`, `Signed`, `SignatureStatus`	BYOVD, rootkits (T1068, T1014)
7	Image (DLL) loaded	`ImageLoaded`, `Image`, `Signed`	DLL hijacking/side-loading (T1574); *very high volume*
8	CreateRemoteThread	`SourceImage`, `TargetImage`, `StartFunction`	Injection (T1055)
9	RawAccessRead	`Device`	Disk read bypassing file ACLs (NTDS.dit, T1003.003)
10	Process access	`SourceImage`, `TargetImage`, `GrantedAccess`, `CallTrace`	LSASS dumping (T1003.001), injection
11	File created	`TargetFilename`, `Image`	Payload drops, staging, ransom notes (T1105, T1074)
12	Registry object created/deleted	`TargetObject`, `EventType`	Persistence setup
13	Registry value set	`TargetObject`, `Details`	Run keys (T1547.001), COM hijack (T1546.015), Defender exclusions
14	Registry key/value renamed		Evasion
15	FileCreateStreamHash	`TargetFilename`, `Contents`	Alternate data streams; Mark-of-the-Web (T1564.004)
16	Sysmon config changed		Tamper (T1562)
17 / 18	Pipe created / connected	`PipeName`, `Image`	C2 named pipes (Cobalt Strike), PsExec-like, lateral movement
19 / 20 / 21	WMI filter / consumer / binding	`Operation`, `Name`, `Query`, `Destination`	WMI persistence (T1546.003)
22	DNS query	`QueryName`, `QueryResults`, `Image`	C2, DGA, tunneling, process-attributed DNS (T1071.004)
23 / 26	File delete (archived / logged)	`TargetFilename`, `Image`	Cleanup (T1070.004); ransomware wipe
24	Clipboard change		Rare; data theft

ID	Event	Key fields	Hunting use / techniques
25	Process tampering	`Type` (Image is replaced / Image is locked for access)	Hollowing, Doppelgänging, herpaderping (T1055)
27	File block executable		Policy enforcement
28	File block shredding		Policy enforcement
29	File executable detected		New PE files written to disk
255	Sysmon error		Health

Useful field conventions

- **`ProcessGuid`** : unique per process instance; use it (not PID) to build process trees.
- **`LogonGuid`** / **`LogonId`** : join to Security 4624.
- **`OriginalFileName`** : from PE resources; survives renaming (e.g. `rclone.exe` renamed to `svchost.exe`).
- **`Hashes`** : pivot with SHA256/IMPHASH; check prevalence across the fleet.
- **`CallTrace`** (event 10): `UNKNOWN(...)` frames often mean injected or reflectively loaded code.
- **`GrantedAccess`** (event 10): `0x1010` (`VM_READ|QUERY_LIMITED`), `0x1410` , `0x1438` , `0x143A` , `0x1FFFFF` (all access); typical credential-dump masks against `lsass.exe` .

Quick SPL / KQL examples

```
index=sysmon EventCode=1 | stats dc(host) as hosts by ParentImage Image | where hosts<3 | sort hosts
index=sysmon EventCode=17 OR EventCode=18 | stats count by PipeName Image | sort count
index=sysmon EventCode=22 | rare limit=20 QueryName
```

```
// Sysmon ingested via Windows Event Forwarding -> Event table
Event
| where Source == "Microsoft-Windows-Sysmon" and EventID == 1
| extend Cmd = extract(@"CommandLine: (.*)", 1, RenderedDescription)
```

Config tips

- Prefer **include-based rules** for noisy events (3, 7, 11, 22), and **exclude-based** for process creation (log nearly everything).
- Log **ImageLoad (7)** only for specific, high-value DLLs or paths, or you will drown.
- Version-control the config; tag the version in a registry key or rule name so you know which hosts run which rules.
- Monitor for Sysmon stopping (event 4, service 7036) and config changes (16).